Machine Learning Applications for Health

健康领域的机器学习应用

COMP90089 (2026) - Lecture 2

COMP90089（2026）- 第 2 讲

Prof Daniel Capurro Daniel Capurro 教授

Today's Agenda 今日议程

- Quick Recap 快速回顾
- Open discussion 开放讨论

What you reviewed before class

课前复习内容

Introduction to Biomedical Informatics

生物医学信息学简介

Learning Healthcare System

学习型医疗系统

What is machine learning?

什么是机器学习？

Sergios Theodoridis
"Machine learning is a name that is gaining popularity as an umbrella and evolution for methods that have been studied and developed for many decades in different scientific communities and under different names, such as statistical learning, statistical signal processing, pattern recognition, adaptive signal processing, image processing and analysis, system identification and control, data mining and information retrieval, computer vision, and computational learning...

“机器学习这一名称正日益流行，成为一个统称，也代表着对数十年来不同科学领域以不同名称研究和发展的方法的演进，例如统计学习、统计信号处理、模式识别、自适应信号处理、图像处理与分析、系统辨识与控制、数据挖掘与信息检索、计算机视觉以及计算学习……

What is machine learning?

什么是机器学习?

“The name “machine learning” indicates what all these disciplines have in common, that is, to learn from data, and then make predictions. What one tries to learn from data is their underlying structure and regularities, via the development of a model, which can then be used to provide predictions.”

“机器学习”这一名称体现了所有这些学科的共同点，即从数据中学习，然后进行预测。人们试图从数据中学习的是其潜在结构和规律，通过构建模型来实现，而该模型随后可用于提供预测。

(Machine Learning: A Bayesian and Optimization Perspective, xiii)

（《机器学习：贝叶斯与优化视角》，第 xiii 页）

Sergios Theodoridis

ML and health informatics

机器学习与健康信息学

Machine learning in health is a part of health informatics (medical informatics, biomedical informatics)

健康领域的机器学习属于健康信息学（医学信息学、生物医学信息学）的一部分

What is informatics? 什么是信息学?

"Informatics is the science of information. It studies the representation, processing, and communication of information in natural and artificial systems. Since computers, individuals and organizations all process information, informatics has computational, cognitive and social aspects.

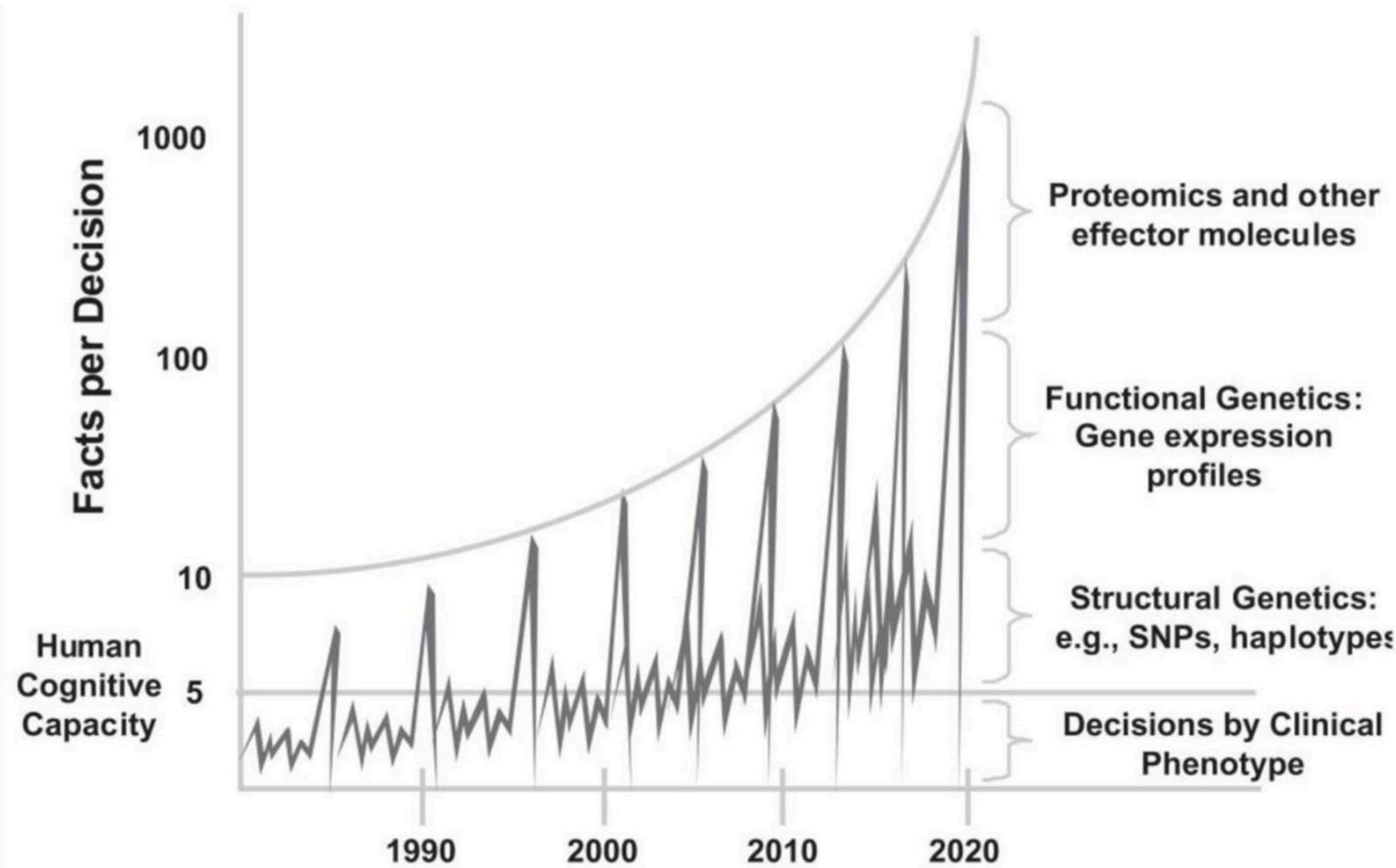
“信息学是信息科学。它研究自然系统和人工系统中的信息表示、处理与通信。由于计算机、个人和组织都会处理信息，因此信息学具有计算、认知和社会层面的内容。

“Used as a compound, in conjunction with the name of a discipline, as in medical informatics, bio-informatics, etc., it denotes the specialization of informatics to the management and processing of data, information and knowledge in the named discipline.”

“当信息学与某一学科的名称结合使用，构成复合词时，例如医学信息学、生物信息学等，它表示信息学在该学科领域中对数据、信息和知识进行管理与处理的专业化。”

Why is health informatics important?

为什么健康信息学很重要?



Why is health informatics important?

为什么健康信息学很重要?

The three numbers you need to know about healthcare: the 60-30-10 Challenge

你需要了解的医疗保健领域三个数字：60-30-10 挑战

Braithwaite et al. BMC Medicine (2020) 18:102

Braithwaite 等。《BMC Medicine》（2020）18:102

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Jeffrey Braithwaite [1*] (e), Paul Glasziou [2] (e) and Johanna Westbrook [3] (e)

Jeffrey Braithwaite [1*] (e)、Paul Glasziou [2] (e) 和 Johanna Westbrook [3] (e)

Abstract 摘要

Background: Healthcare represents a paradox. While change is everywhere, performance has flatlined: 60% of care on average is in line with evidence- or consensus based guidelines, 30% is some form of waste or of low value, and 10% is harm. The 60-30-10 Challenge has persisted for three decades. Main body: Current top-down or chain-logic strategies to address this problem, based essentially on linear models of change and relying on policies, hierarchies, and standardisation, have proven insufficient. Instead, we need to marry ideas drawn from complexity science and continuous improvement with proposals for creating a deep learning health system. This dynamic learning model has the potential to assemble relevant information including patients' histories, and clinical, patient, laboratory, and cost data for improved decision-making in real time, or close to real time. If we get it right, the learning health system will contribute to care being more evidence-based and less wasteful and harmful. It will need a purpose-designed digital backbone and infrastructure, apply artificial intelligence to support diagnosis and treatment options, harness genomic and other new data types, and create informed discussions of options between patients, families, and clinicians. While there will be many variants of the model, learning health systems will need to spread, and be encouraged to do so, principally through diffusion of innovation models and local adaptations. Conclusion: Deep learning systems can enable us to better exploit expanding health datasets including traditional and newer forms of big and smaller-scale data, e.g. genomics and cost information, and incorporate patient preferences into decision-making. As we envisage it, a deep learning system will support healthcare's desire to continually improve, and make gains on the 60-30-10 dimensions. All modern health systems are awash with data, but it is only recently that we have been able to bring this together, operationalised, and turned into

useful information by which to make more intelligent, timely decisions than in the past.

背景：医疗保健领域存在一个悖论。尽管变化无处不在，但绩效却陷入停滞：平均而言，只有 60%的医疗服务符合基于证据或共识的指南，30%属于某种形式的浪费或低价值服务，10%会造成伤害。“60-30-10 挑战”已经持续了三十年。

正文：目前，自上而下或遵循链式逻辑的应对策略，本质上基于线性变化模型，并依赖政策、层级和标准化，但事实证明这些策略并不足够。相反，我们需要将复杂性科学和持续改进的理念，与构建深度学习型医疗卫生系统的方案结合起来。这种动态学习模型有望汇集包括患者病史，以及临床、患者、实验室和成本数据在内的相关信息，从而在实时或接近实时的情况下改善决策。如果我们能够正确实施，学习型医疗卫生系统将有助于使医疗服务更加循证，并减少浪费和伤害。 它需要专门设计的数字骨干和基础设施，应用人工智能来支持诊断和治疗方案，利用基因组数据及其他新型数据，并促成患者、家属与临床医生之间就各种方案展开知情讨论。尽管这一模式会有许多变体，但学习型卫生系统需要推广开来，并且主要应通过创新扩散模式和本地化调整来鼓励其发展。结论：深度学习系统能够帮助我们更好地利用不断扩展的健康数据集，包括传统数据以及较新形式的大规模和小规模数据，例如基因组数据和成本信息，并将患者偏好纳入决策。如我们所设想的那样，深度学习系统将支持医疗卫生领域持续改进的愿望，并在 60-30-10 这三个维度上取得进展。 所有现代卫生系统都充斥着数据，但直到最近，我们才有能力将这些数据汇集起来，加以运用，并转化为有用的信息，从而比过去做出更明智、更及时的决策。

Keywords: Learning health system, Complexity, Complexity science, Change, Evidence-based care, Clinical networks, Quality of care, Patient safety, Policy, Healthcare systems

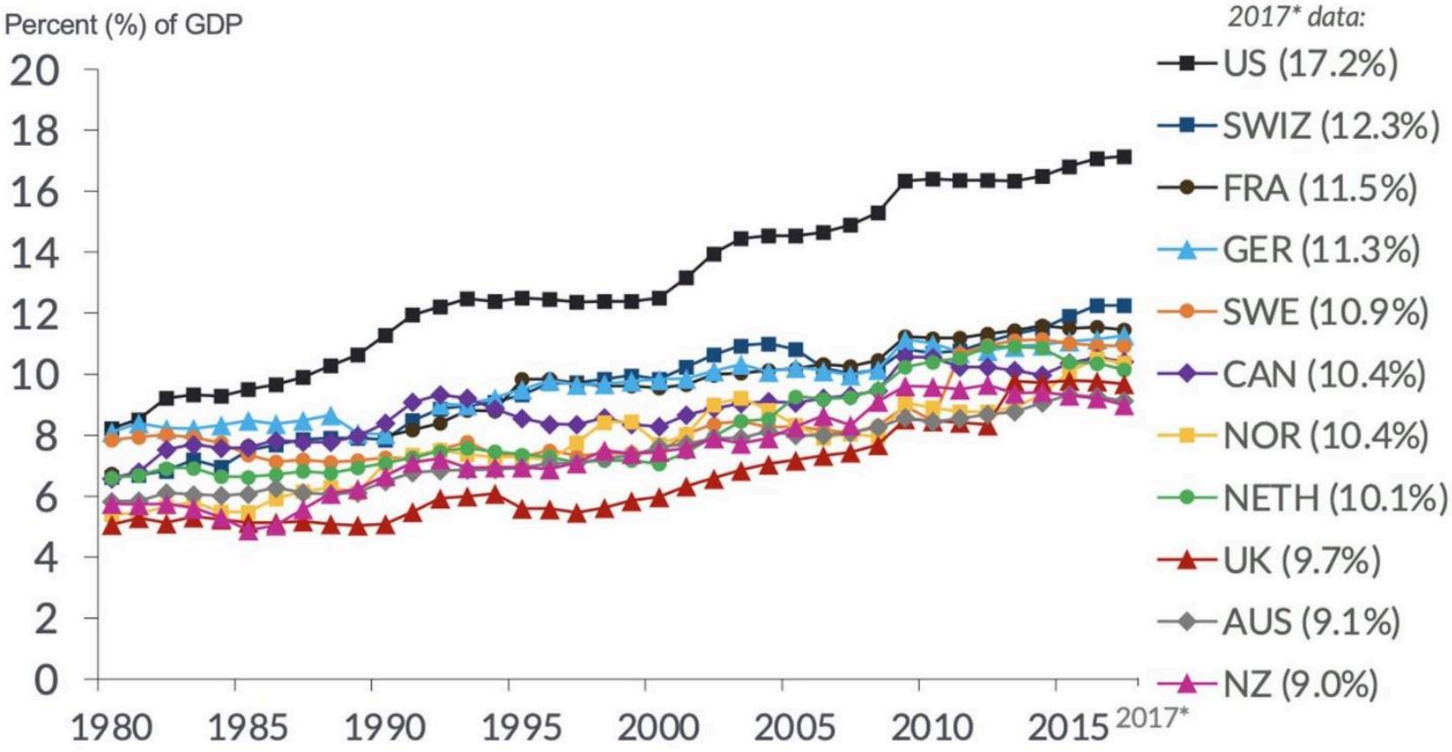
关键词：学习型医疗卫生系统、复杂性、复杂性科学、变革、循证医疗、临床网络、医疗质量、患者安全、政策、医疗卫生系统

Health Care Spending as a Percent of GDP, 1980-2017

医疗卫生支出占 GDP 的百分比，1980—2017 年

Adjusted for Differences in Cost of Living

已根据生活成本差异进行调整



Key Concepts 核心概念

What informatics "is and isn't" (Friedman, 2013)

信息学“是什么与不是什么”（Friedman，2013）

- Is 是
  - Cross-training where basic informational sciences meet a biomedical application domain

    基础信息科学与生物医学应用领域相结合的交叉培训

  - Relentless pursuit of assisting people

    坚持不懈地致力于帮助人们

  - Tower of achievement 成就之塔
    - Model formulation 模型构建
    - System development 系统开发
    - System implementation 系统实施
    - Study of effects 效果研究
- Isn’t 难道不是
    - Scientists or clinicians tinkering with computers

      科学家或临床医生摆弄计算机

    - Analysis of large data sets per se

      大数据集本身的分析

    - Circumscribed roles related to deployment of electronic health records (*point of disagreement)

      与电子健康记录部署相关的限定性角色（*争议点）

    - Profession of health information management

      健康信息管理专业

  - Anything done using a computer

    使用计算机完成的任何事情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↓ 10.6%
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Quality Score
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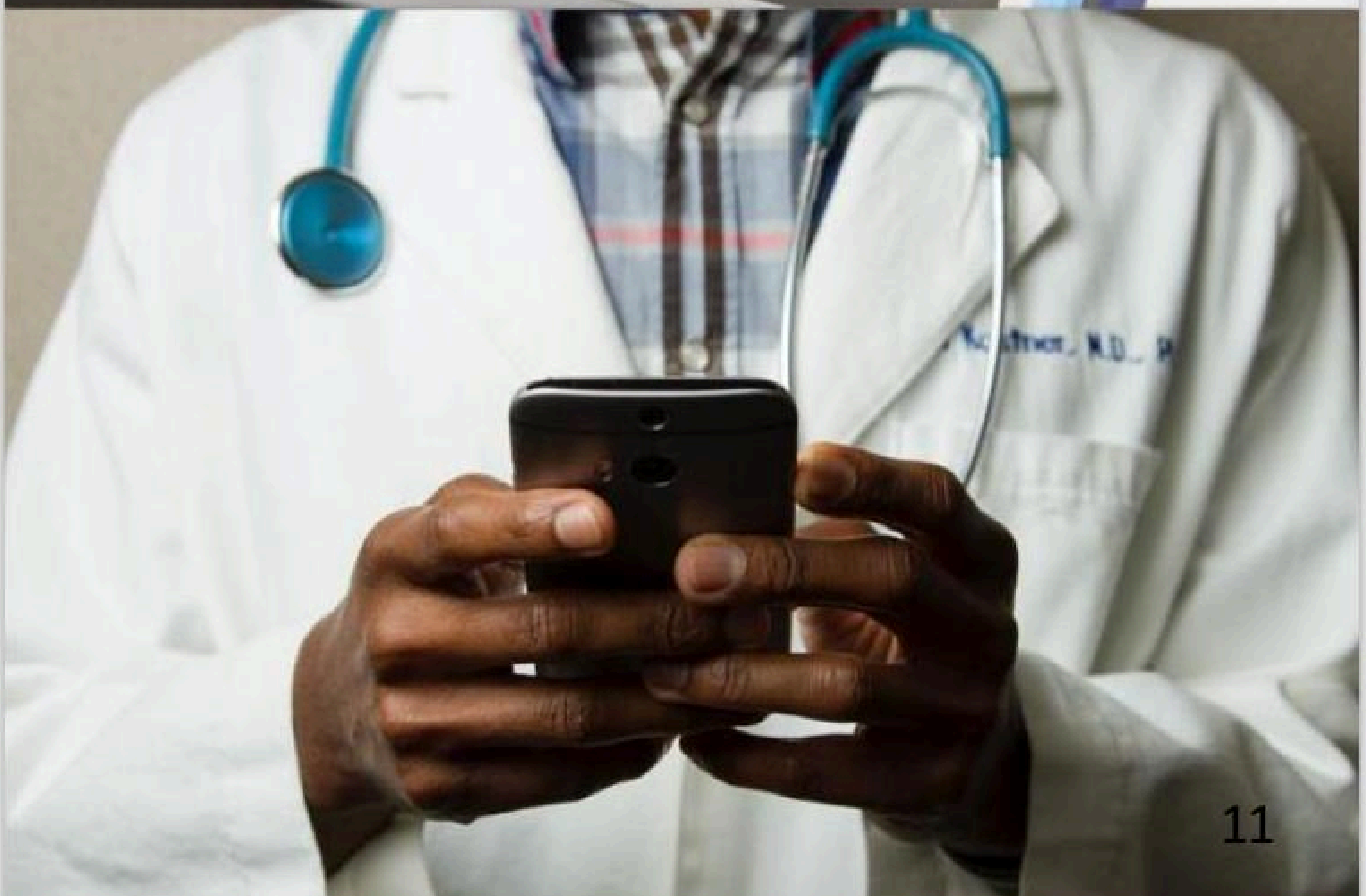
↓ -0.1%

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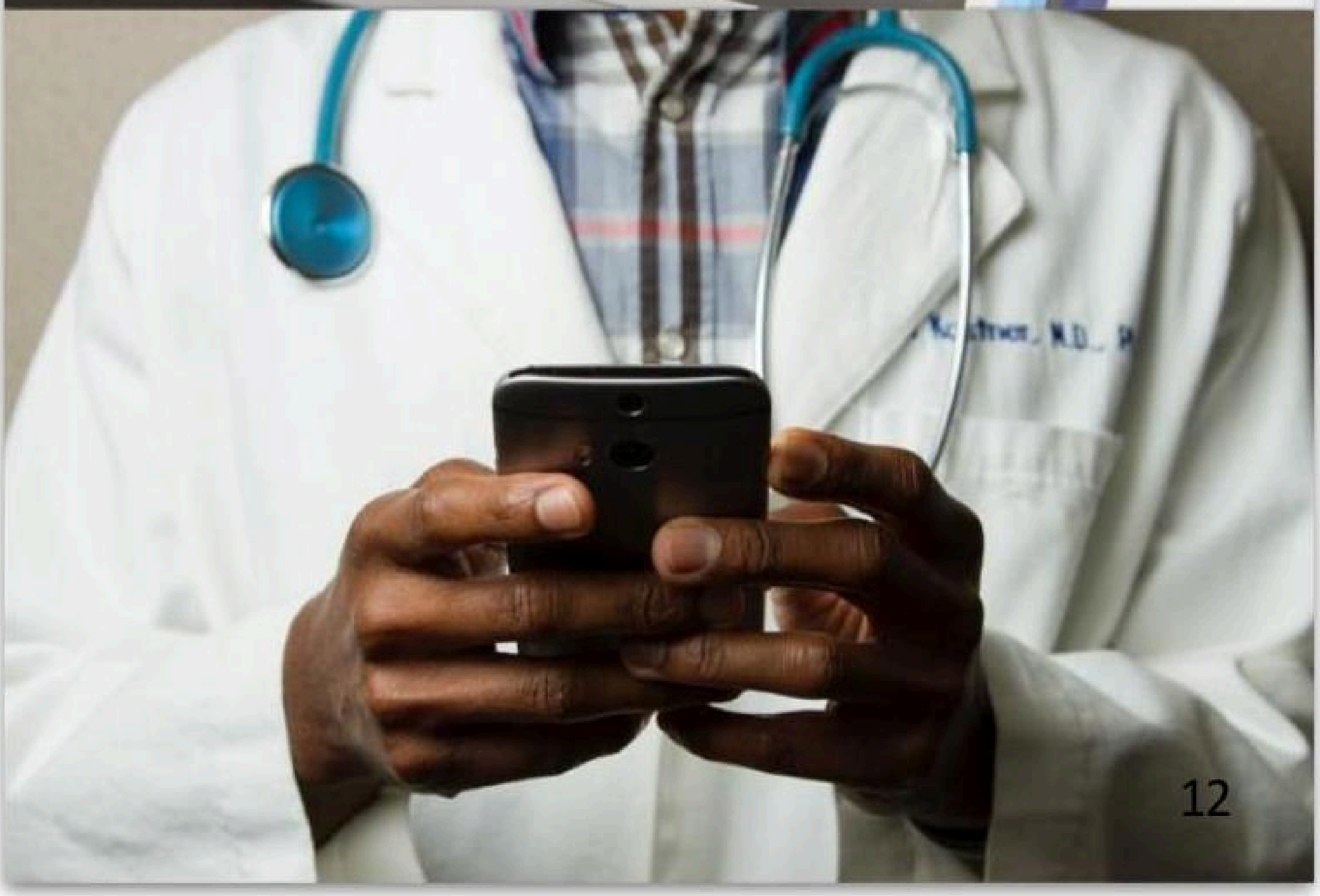

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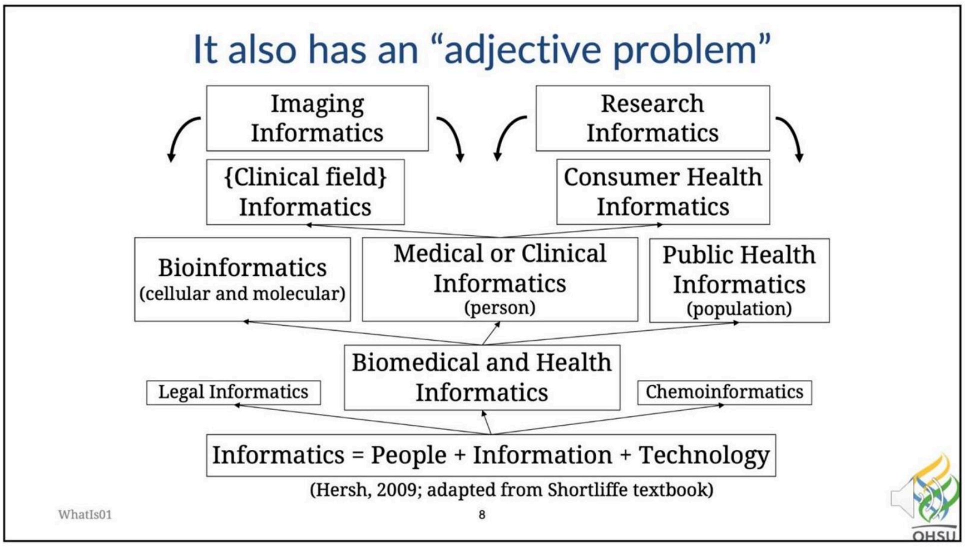

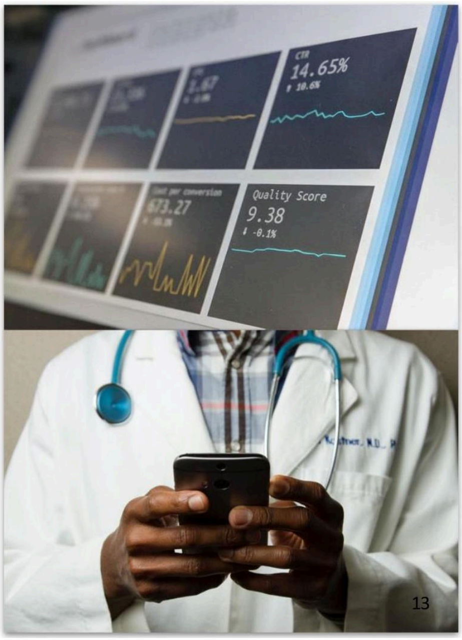

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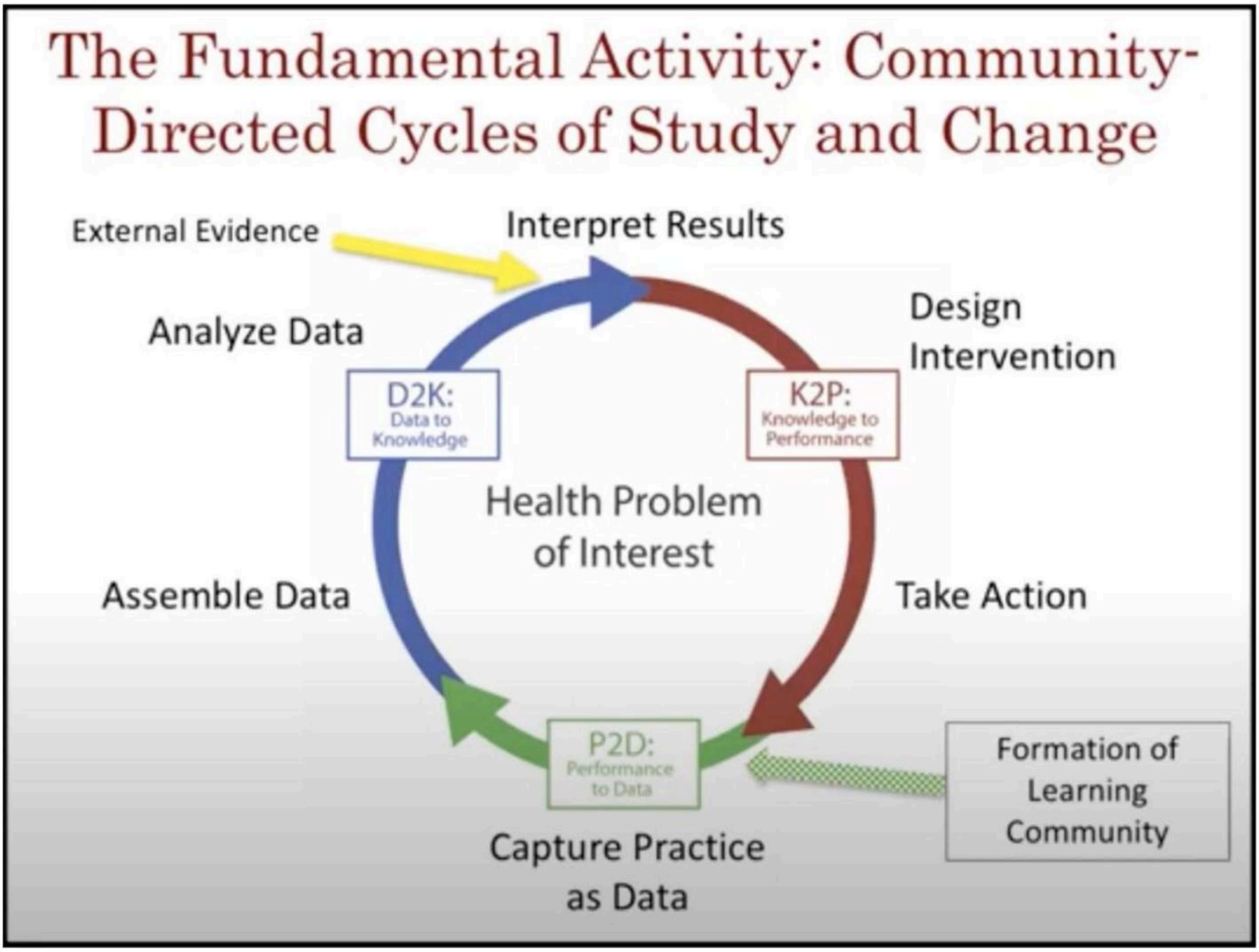


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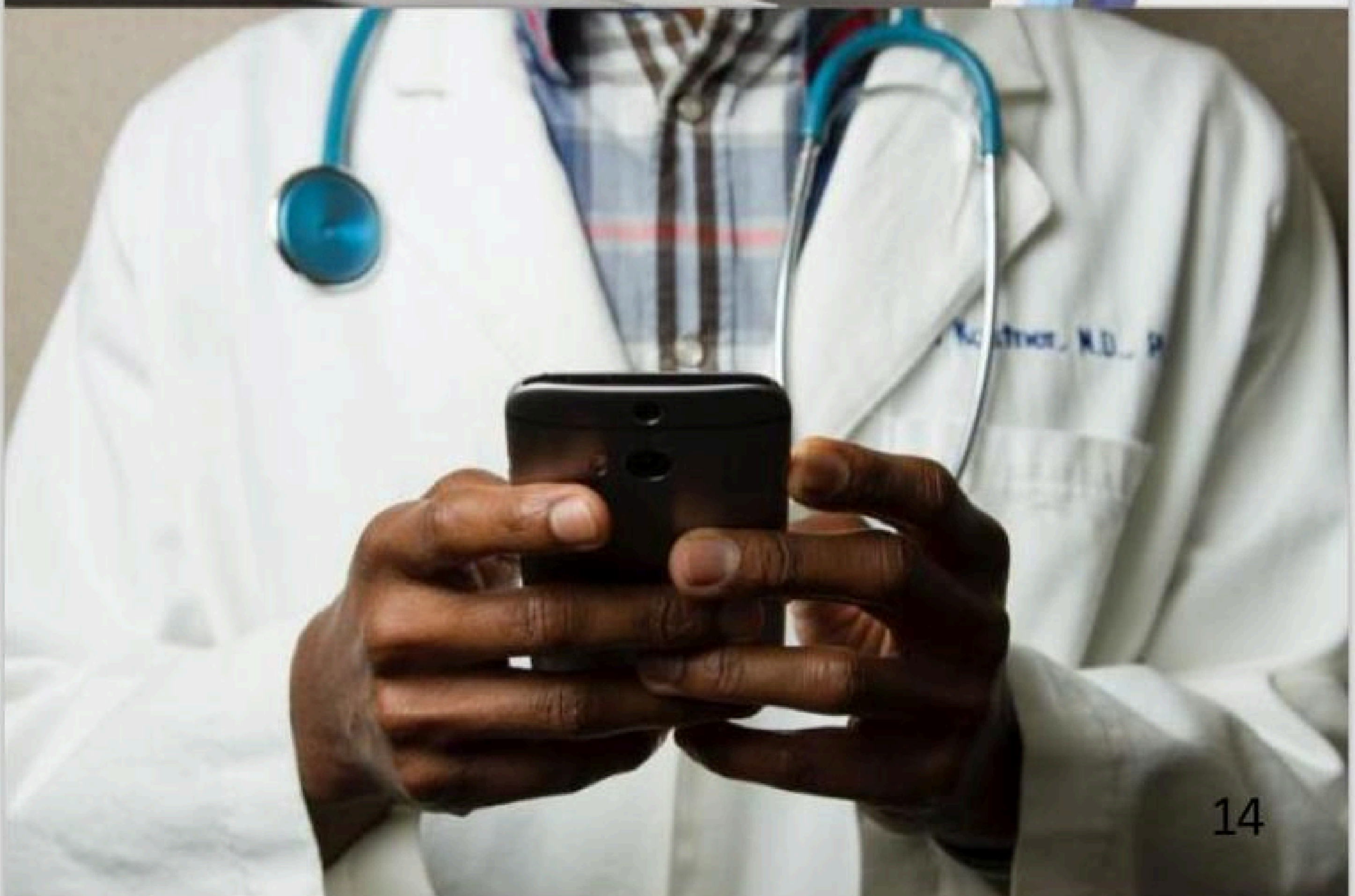


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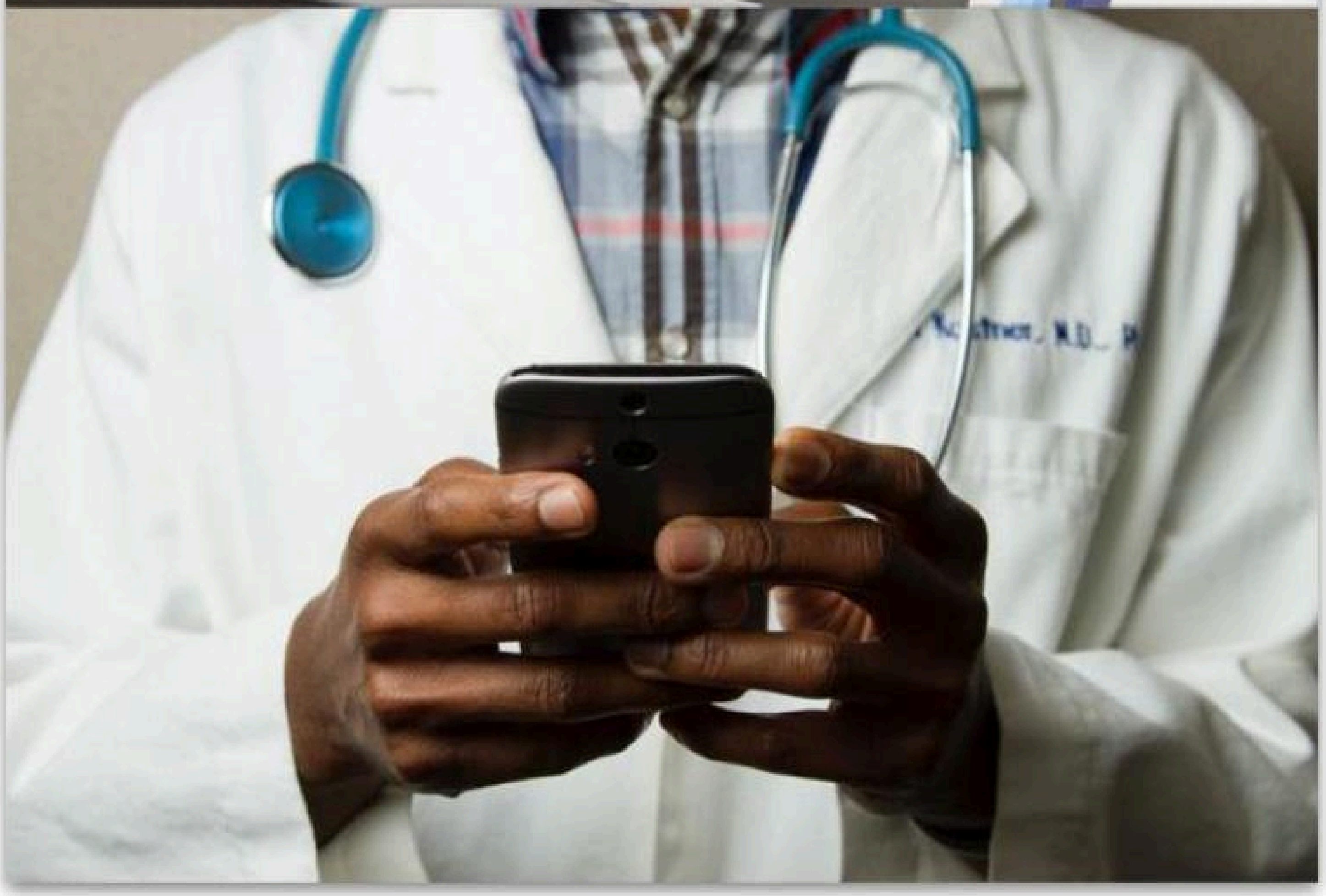

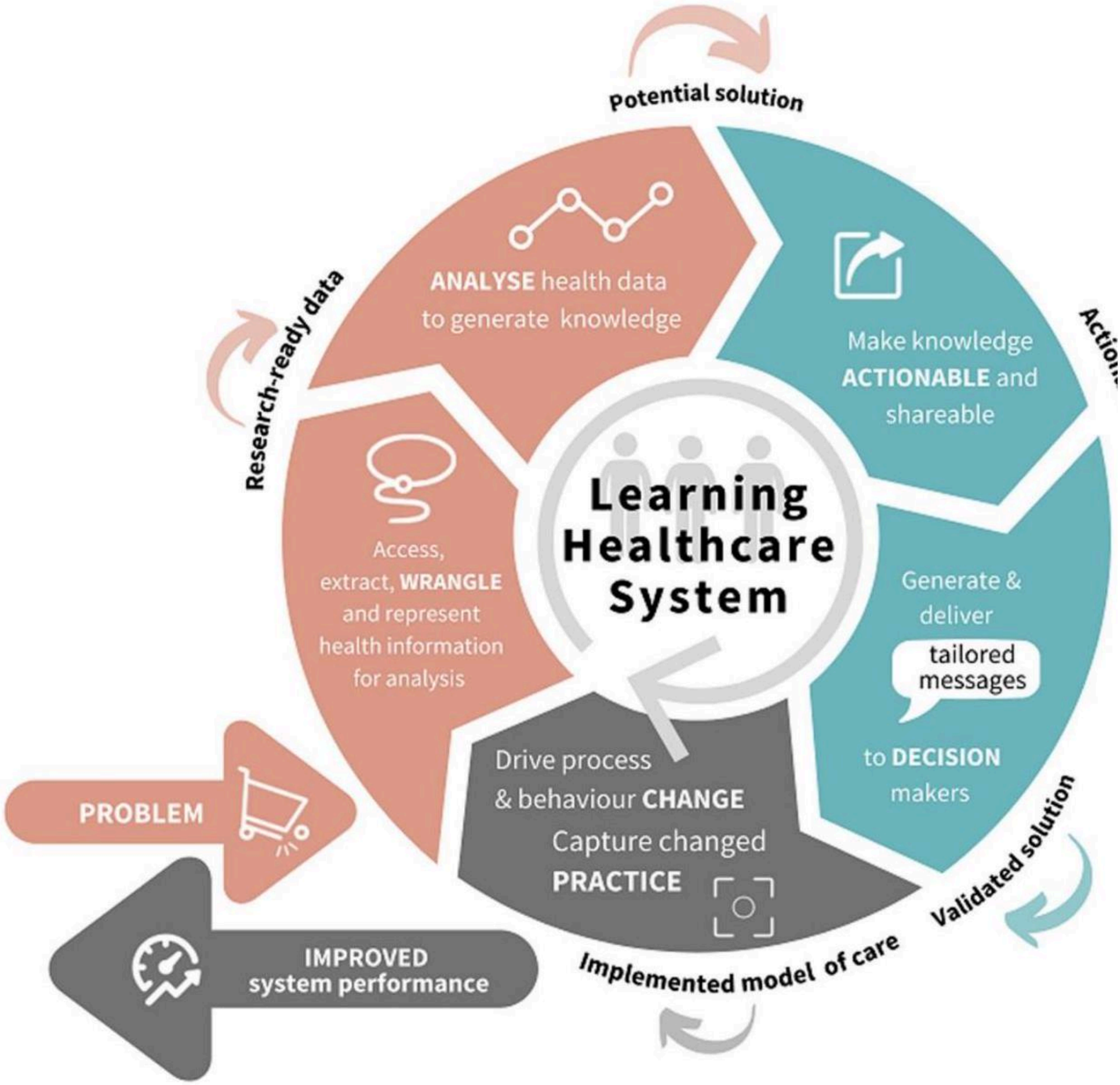

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